Experimental Studies on a Low Power Injection-Locked Continuous Wave Magnetron

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Abstract — An injection-locked magnetron was investigated at low power injection. The 1 kW magnetron has been locked over the ratio of the input power to output power of -57 dB. It is much less than the required injection power in previous experiments. The purity of spectrum was presented. The phase noise of the locked magnetron was less than -93.1 dBc/Hz at 10 kHz offset. The oscillation spectrum has FM noise due to the spurious of input reference signal and injection locking contributes to reducing the FM noise itself to some extent. The injection locking technique may be applied to power combining or large-power amplifiers based on magnetrons.

Index Terms—Magnetron, injection locking, power amplifier, noise

I. Introduction

The majority of existing frequency selective heating [1], communication transmitters and SAR radar utilizes technologies such as traveling wave tube amplifier (TWT) [2], klystrons [3] and GaN based solid state power amplifier (SSPA) for the power amplifier [4]. Recently, Kwack et al. realized a GaN power amplifier which delivered more than 1 kW and a typical 34% PAE [4]. Song et al. proved a compact and lightweight S-band 1 kW traveling-wave tube with 25 dB saturated gain and over 40% efficiency [2]. However, both of them are much too expensive for industrial applications and have a high cost at per watt of output power because of efficiency.

Magnetrons have been widely used in industrial heating systems because of their low cost and low consume of energy for per watt of output power. Unfortunately, microwave power applications are expected to stabilize the frequency as well as supply power. Magnetrons, whose amplitude and phase could not be accurately controlled, are widely regarded as noisy oscillators. To effectively solve the aforementioned problem of magnetrons, injection locking is the key technology to achieve frequency and phase control. Mitani et al. achieved a phase-controlled magnetron whose phase noise estimated to be less than -85 dBc/Hz at 10 kHz for a microwave power transmission system of solar power satellites, was locked about the ratio of the input power to output power of -30 dB [5]. Tahir et al. demonstrated utilizing the magnetron frequency pushing characteristic, a 2.45 GHz ‘‘cooker’’ magnetron locked by 40 dB injection signal below the output power [6]. The research and related technology on magnetron are still continued to obtain the lower power of injected signal and/or the purer spectral purity of magnetron. Therefore, understanding the relationship between the locked magnetron’s output and injection signal, considered as pulling characteristic, is essential in injection-locking. Moreover, the actual locked magnetron presents some nonlinear behaviors and no such studies have been conducted to date.

This paper presents an injection-locked magnetron using a low power of reference signal, an analysis of the relationship between the locked magnetron’s output and injection reference signal, and an investigation on spectral purity of the magnetron locked by the ratio of the input power to output power of -57 dB.

II. Experiment

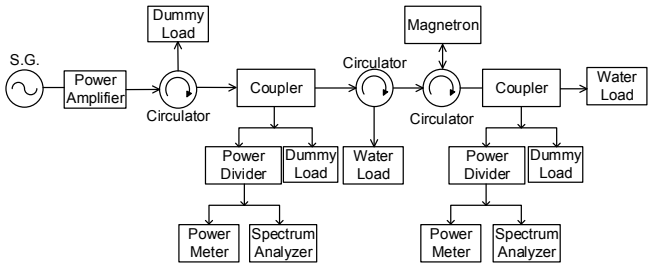


Fig. 1 Block diagram of the low power injection-locked CW magnetron system

The block diagram of 1 kW continuous-wave injection-locked magnetrons microwave measurement system is shown in Fig. 1. For the magnetron, 2.45 GHz oven magnetron 2M210 - M1 made by Panasonic was used. The driving power supply for the magnetron was a self-designed DC analog stabilized power supply without any special bias voltage control which had a ripple of less than 1.5‰. The injection signal was provided by the vector signal generator (E8267C, Agilent) amplified with the 50 dB gain amplifier (ZHL-30W-262-S+,

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Mini-Circuits). The three-port circulators were applied to protect the magnetron, amplifier and guide the injected signal to magnetron. Two three-port circulators in series were to achieve high isolation. The microwave output coupled from the magnetron to a waveguide was absorbed by a high-power water load (WR430 high-power water load, Euler, Chengdu). The input power of the injection signal and the output power of the load were measured by a power meter (AV2433, the 41st Institute of CETC). The frequency and phase noise of the injection signal and magnetron were measured by a signal analyzer (FSV40, Rohde & Schwarz).

The center frequency of free-running magnetron oscillated at 2.4510 GHz with anode voltage -3603 V and anode current 0.36 A, after oscillation the cathode current was turned off to reduce the FM noise and stabilize the oscillation with a narrowband spectrum [5].

III. Results and Discussion

An injection-locked magnetron is a magnetron locked by an injected reference signal. The standard theory assumes that the circuit of Adler's equivalent model operates as a linear amplifier and the external reference signal should be weak, so the locking magnetron output and the external reference signal could be satisfied with the well-known Adler's condition [7] as follow.

$$P_{ref} \ll P_{out} \tag{1}$$

$$\sin\alpha = 2Q\sqrt{\frac{P_{out}}{P_{ref}}}\bullet\frac{\Delta\omega}{\omega} \tag{2}$$

where α , $\Delta\omega$ are phase and frequency difference between the reference signal and the output of free-running magnetron respectively, Q is loaded quality factor of the magnetron, ω is the instantaneous angular frequency of the magnetron, P_{ref} is injected power, P_{out} is output power, P_{ref}/P_{out} is defined as the injection power ratio. Thus, the behaviors of magnetron is expected to be improved by injection locking. The relationship between the reference signal and injection-locked magnetron output is discussed.

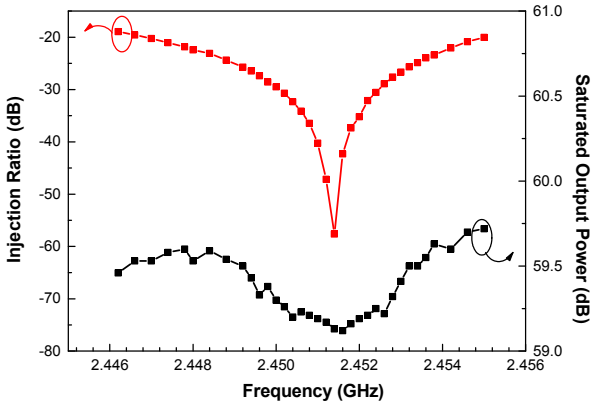


Fig. 2 The minimum injection power ratio and saturated output power at different frequency

The injection power ratio at each frequency and magnetron saturated output power was measured by pulling the input signal to the extremes of the locking range, the results were listed in Fig. 2. The injection power ratio increased with frequency shift away from the center frequency and output power from the magnetron was approximately constant, it was apparent that the magnetron saturated output power positively correlated with injection ratio. It should point out that without any rigorous restrictions the magnetron could be locked over the ratio of the input power to output power of -57.6 dB at 2.4514 GHz near the center frequency of a free-running filament-off magnetron and the DC to RF efficiency of magnetron was also measured more than 65.9%. The low noise performance of high-power oscillators and power supply ripples are the key factor for phase locked easily [8]. Besides, the magnetron was still full locked or locked within a period of time by further decreased the power of injection signal which was not swamped by the reflections. These phenomena would be discussed in our further studies.

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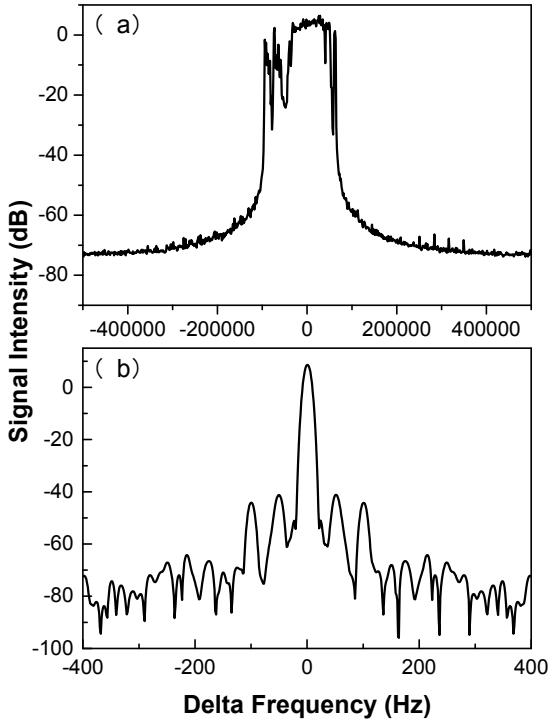


Fig. 3 The oscillation spectrum of magnetron at different states (RBW: 10Hz, VBW: 10Hz)

Fig. 3 exhibited the close-up view of free-running magnetron (max hold) and locked frequency (2.451 GHz) characteristics of output power to the load with the ratio of the input power to output power of -57 dB. Both frequency and amplitude of the free-running magnetron could not be stable, as shown in Fig. 3 (a). Neither a phase locked loop acting on the anode current nor superconducting cavity used in system, The center spectral width of oscillation output was less than 100 Hz and it could be obviously seen that - 41.2 dB microphonic (spurious suppression equal to -49.8 dBc) at ±50 Hz in Fig. 3 (b), which was caused by the ripple of the analog stabilized power supply. The maximum peak-to-peak amplitude fluctuation was ±0.2 dB and the obvious improvement of output phase noise was measured to be less than -93.1 dBc/Hz at 10 kHz, compared to Mitani's work [5].

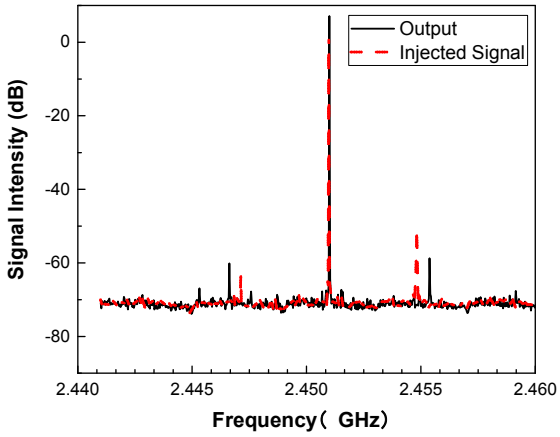


Fig. 4. FM noise suppression in the injection locking operation (RBW: 30Hz, VBW: 30Hz)

Fig. 4 presented the spectrum of input injection signal and output of locked magnetron. The center frequency (2.4510 GHz) was perfectly matched indicated that the output of magnetron was injection-locked. An apparent spurious spectrum at 2.4548 GHz (spurious suppression equal to -52 dBc) appeared in the injection signal led to two mirror frequency spurious signal occurred in output and the frequency of upper output spurious signal was shift to 2.4553GHz. The improved suppression of both output spurious signal was - 66 dBc. In other words, the spurious of input reference signal could generate FM noise as well, it appears nonlinear unlike the linear power amplifier, but it is still attractive because the injection-locking can reduce some FM noise itself.

To the best of our knowledge, the achieved injection ratio is much less than the required injection power in previous experiments. For a better evaluation, a comparison is given among the documented previous designs (Table I). Based on these pulling characteristics, the injection-locked magnetron has promising potentials to work as a cheap but high gain large power amplifier in industrial or radio astronomy application. Moreover, we have found that modulations may be applied to the injection locked magnetron.

TABLE I
COMPARISON AMONG PREVIOUSLY DESIGNED INJECTION-LOCKED MAGNETRONS

Source	Output Power (W)	Injection ratio (dB)	Phase noise	Spurious Suppression
[5]	—	-30	-85dBc/Hz at 10 kHz	—
[6],[8]	1000	-40	—	-13dBc at 100 Hz

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				/ -35dBc at 40 kHz
[9]	825	-22	—	-15 dBc at 3 MHz
[10]	380	-26	-70dBc/Hz at 40 kHz	—
[11][12]	530	-30	-59dBc/Hz at 59 Hz	-37 dBc at 31 Hz
This work	⩾900	-57	-93dBc/Hz at 10 kHz	-49.8dBc at 50 Hz

IV. Conclusion

An injection-locked magnetron was investigated at a low power injection. The magnetron was locked over the ratio of the input power to output power of -57 dB and the purity of spectra was presented. The phase noise of the locked magnetron of was less than -93.1 dBc/Hz at 10 kHz offset. The oscillation spectrum has FM noise due to the spurious of input reference signal and injection locking contributes to reducing the FM noise to some extent. Also, the harmonic suppression behavior of injection-locked magnetron was demonstrated. The injection-locked magnetron has promising potential to work as a cheap and high gain power amplifier in industrial application or radio astronomy. Moreover, the technique will find applications in power combining based on magnetrons, from which a high power and low cost microwave source will be built for industrial applications.

Acknowledgments

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